MCO-1D

Operator Manual

How to fly, talk to, and work with your ornithopter drone.

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Revision	23 — August 2026
New in v23	Standard six-pack instrument layout · Reference bugs (VSI / heading / altitude) · Precision flying · Stroke-smoothed
Audience	Pilot / Operator — no engineering background required
Companion	Commissioning Manual v20 · Wing Addendum R21

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1 What This Manual Is For

This manual covers everything you need to fly the MCO-1D ornithopter drone. No engineering knowledge required. You fly it like any aircraft.

◆ **NOTE** Every layer of software is completely invisible. You move the controls. The aircraft responds. The reference bugs show you where to aim — you fly there.

Aircraft at a Glance

Aircraft	MCO-1D metachronal cascade ornithopter SAR drone
Wings	8 — four tandem pairs, independently controlled
Weight	~4.1–5.1 kg flying
Stall	4.85 m/s (~9.4 kt)
Max speed	28 m/s (~54 kt)
Instruments	Standard six-pack + power/battery in status strip
Interface	Plain English conversation

2 Equipment

Aircraft side: commissioned MCO-1D, charged batteries, clear area.

Ground station:

- Laptop — any modern browser
- Python 3.9+ with `pip install aiohttp anthropic`
- Anthropic API key — set as environment variable `ANTHROPIC_API_KEY`
- Files: mco_ws_bridge.py · mco_api_proxy.py · mco_ground_station_launcher.sh
- Interface: mco_ai_ground_station.jsx in Claude Artifacts

> ■ **CAUTION** Keep your API key in an environment variable only. Never in a file.

3 Starting the Ground Station

Power aircraft → connect laptop to aircraft network → set API key → run launcher → open interface → check connection badge.

```
export ANTHROPIC_API_KEY=sk-ant-...
bash mco_ground_station_launcher.sh 192.168.7.1
```

■ **CAUTION** Green DISARMED badge = ready. Red OFFLINE = check launcher and aircraft.

4 The Instrument Panel — Six-Pack Layout

The ground station displays the standard six-pack instrument arrangement familiar to every fixed-wing pilot, with power plant indicators in the status strip below.

TOP ROW	BOTTOM ROW
Airspeed indicator	Turn coordinator
Attitude indicator	Directional indicator (heading)
Altimeter	Vertical speed indicator (VSI)

Status strip (below six-pack)

Armed / Safe	Arming state indicator. Blinks amber when armed.
Faults	Fault count. Green OK or red count with blinking.
State	Current flight state — cruise, hover, ALSS phase, etc.
Discovery	Learning system active indicator.
PWR	Motor power in kilowatts. Green < 3 kW, amber 3-4.2, red > 4.2.
BAT	Battery voltage. Green > 23V, amber 21.5-23V, red < 20V.
HDG■	Heading bug: target heading and error. Shown only when bug is set.
ALT■	Altitude bug: target altitude and error. Shown only when bug is set.
ALSS	Autonomous learning sortie progress. Shown only during ALSS.
Ceiling	Ceiling probe altitude and LEV risk. Shown only during probe.

Gauge details

Airspeed	0–30 m/s. Red arc 0–4.85 (below stall). Green 4.85–20. Amber 20–26. Red 26–30 (Vne region).
Attitude	Blue sky / brown earth split with bank arc. Shows pitch and roll simultaneously.
Altimeter	Two needles: main (0–50m per revolution), minor (0–500m total). Altitude bug shown as amber triangle
Turn coordinator	Miniature aircraft banks to show turn rate. Inclinometer ball shows slip/skid. Index marks at standard ra
Directional indicator	Rotating compass card. N marks the 12 o-clock position. Heading bug shown as amber triangle on the
VSI	–5 to +5 m/s. Centre zero. Positive up (climbing). Display is stroke-smoothed over 5 wingbeats. VSI bu

◆ **NOTE** The VSI reads the stroke-smoothed vertical speed — the five-wingbeat rolling average. The raw flapping cycle produces force variation at 3.74 Hz which would make the needle bounce ±82 ft/min continuously. The averaging removes this entirely. What you see is the true mean rate.

5 Reference Bugs — VSI, Heading, Altitude

Three reference bugs are available, one on each primary flight instrument. All three work identically: you set a target, the instrument shows it, you fly to it. The aircraft does nothing.

◆ **NOTE** The bugs are reminders, not commands. Setting a heading bug does not make the aircraft fly that heading. Setting a 100 ft/min descent bug does not make the aircraft descend at 100 ft/min. You set the power and pitch to put the needle on the bug. You monitor it. You correct it. You are flying the aircraft.

VSI bug — vertical speed reference

The VSI gauge always shows an amber triangle at the commanded rate. Default is zero (level flight). The readout shows current rate above the reference value. Green within 50 ft/min, amber within 150, red outside.

"target 100 ft/min down / descent rate 100"
→ vsi_ref -100. Amber triangle moves to -100 ft/min on gauge.

"500 ft/min descent"
→ vsi_ref -500.

"200 ft/min climb"
→ vsi_ref 200.

"level flight / clear the VSI bug"
→ vsi_ref 0. Triangle returns to centre.

Heading bug — directional indicator reference

Amber triangle appears on the rotating compass card. Status strip shows target heading and signed error. Green within 5 degrees, amber within 15, red outside.

"set heading bug to 090 / fly heading 090 / target heading east"
→ hdg_ref 090. Triangle appears on DI card at 090 degrees.

"heading bug north"
→ hdg_ref 360.

"clear the heading bug"
→ hdg_ref_clear. Triangle removed from card.

Altitude bug — altimeter reference

Amber triangle on the altimeter minor needle scale (0-500m). Current altitude shown in green when within 5m of bug, amber within 20m, red outside. Accepts feet or metres.

"altitude bug 1000 feet / set the altimeter bug to 1000"
→ alt_ref 1000. Triangle appears at 1000 ft (305m) on altimeter.

"300 metres AGL"
→ alt_ref 300 m.

"clear the altitude bug"
→ alt_ref_clear.

◆ **NOTE** All three bugs persist until cleared. They do not reset between flights or when the mode changes. GS: precision_status shows all three bugs and current errors in one report.

6 The Conversation Interface

Type plain English. The AI understands intent, operates the aircraft, and explains results.

Six-pack (top)	Six flight instruments in standard layout.
Status strip	Armed state, faults, state, PWR, BAT, and any active bugs.
Conversation	Your messages right, AI replies left. Red for alerts.
Input box	Type and press Enter.

◆ **NOTE** The AI watches the instruments. It speaks up about faults, low battery, and any condition needing attention.

7 Pre-Flight Checks

"Ready to fly"

→ AI checks faults, battery, attitude hold, airspeed sensor, GPS, wing guard calibration.

■ **CAUTION** Wing guard requires fuselage width and TPU start measured and confirmed. AI will tell you what to measure if not done.

Arming

"Arm it" → AI confirms once → arms → badge turns amber. Stand clear of all eight wings once armed.

8 Normal Flight Operations

HOVER	Lever = altitude. Stick = lateral. Release = hold position.
FIXED-WING	Lever = thrust/braking. Stick = bank/pitch/yaw.

◆ **NOTE** Say "take off", "land", "hold position" at any time.

9 The Lever — Complete Guide

Forward = more power. Back = less power and braking. 50% = altitude hold.

ZONE	LEVER	WHAT HAPPENS
THRUST	0–5%	Maximum burst / sprint.
SPRINT	5–28%	High-speed sprint.
POWERED CRUISE	28–45%	Normal cruise.
ALTITUDE HOLD	45–55%	Holds altitude. 50% = exact.
GLIDE / RANGE	55–75%	Efficient glide.
LIFT DUMP PARTIAL	75–80%	Rear wings reduce lift.
LIFT DUMP FULL	80–87%	Rear wings stop — full spoilers.
SHUTTLE BRAKE ½	87–93%	Progressive aerodynamic braking.
J-HOOK	93–97%	Post-stall pitch-up to hover (>20m).
SHUTTLE BRAKE FULL	97–100%	Maximum drag. ~2g.

10 Braking and Deceleration

Lift dump (75–87%)	Rear wings reduce/stop → steeper approach.
Shuttle brake (87–97%)	All wings to drag. 1.5–2g. Auto-exits.
J-hook (93–97%, >20m)	Post-stall pitch-up → hover in <3s.
Emergency stop	Cuts all motors. Only if crash is certain.

■ **IMPORTANT** Emergency stop is not a landing method.

11 How the Aircraft Learns

The aircraft improves continuously. Every flight adds to its knowledge.

Discovery	Tests configuration variations during stable cruise. Runs automatically.
Archive	Stores best-ever config for every airspeed × wind × intent combination.
Strategy genome	Learns which wing mechanisms work best for each manoeuvre.

"That worked well"

→ Positive note recorded. Configuration gets credit.

"The hover feels wobbly"

→ AI adjusts focus to stability.

■ **CAUTION** Say "we're done" after landing to save session learning.

12 Mission Focus

Tell the aircraft what to get better at.

WHEN YOU SAY…	WHAT IT DOES
"Fast / top speed"	Maximum speed pursuit.
"Get there quickly"	Efficient high-speed transit.
"Fastest climb"	Best rate of climb.
"Clear the trees"	Steepest climb angle.
"Aerobatics / tight turns"	Roll authority and precision.
"Precision flying / fly steady"	Trim quality — lowest hands-off variation.
"SAR / rescue"	SAR hover and confined-space ops.
"Stormy conditions"	Storm resilience.
"Stay up longer"	Endurance — minimum power.
"Enable clap and fling"	Hover lift enhancement (~25%).
"Back to normal"	General optimisation resumes.

13 Wing Coordination and Clap-and-Fling

The wing guard is always active, invisible, and cannot be disabled. Clap-and-fling gives ~15-25% more hover lift with the gate open.

"Enable clap and fling"
→ Opens clap gate. Hover lift improves.

"Disable clap / normal wings"
→ Closes clap gate.

"Anti-phase cruise"
→ 115° amplitude. ~20% more stroke volume.

■ **CAUTION** Fuselage width and TPU start must be measured and confirmed before first flight.

14 High-Speed Performance

Three speed regimes — all transitions automatic, all invisible to pilot.

0–15 m/s	**Efficient cruise**	Strouhal-optimal. Best fuel efficiency.
15–22 m/s	**Sprint**	Fixed-wing morphing. Rear pairs generate lift free.
22–28 m/s	**Maximum speed**	Drag minimisation. Maximum sweep.

"Top speed / as fast as possible"
→ Top speed focus.

"Get there quickly / fast transit"
→ Fast cruise focus.

15 Climb Performance

Climb mode is selected invisibly from throttle + pitch combination. Same inputs a conventional fixed-wing pilot uses.

Best rate (high lever + moderate aft)	Fastest altitude gain. ~9-10 m/s.
Best angle (high lever + hard aft)	Steepest slope. ~6-8 m/s.
Sustained (high lever + gentle aft)	Thermal-limited. Hours of climb.
Zoom (aft pull from >15 m/s)	Trades speed for altitude.

"Fastest climb"
→ Best rate focus.

"Clear the trees / steep departure"
→ Best angle focus.

16 Aerobatic Flight

The aircraft reads your stick inputs and routes to the correct aerobatic configuration silently. No mode to engage. Fly it.

WHAT YOU DO	WHAT IT IS
Full roll + aft pitch, fast throw	Snap roll
Full roll, fast throw, no pitch	Aileron roll
Roll + opposing yaw	Lag roll
Roll + slight constant aft	Barrel roll
Roll + firm aft + high lever	Chandelle
Roll + firm aft, any lever	Wingover
Fast throw + full roll + aft + V>12 + lever>70%	Immelmann
Roll established, then push forward	Split-S
Slow deliberate full deflection	Slow roll

Stick urgency — how fast you throw — determines snap vs aileron roll. Each manoeuvre has its own response time: snap roll fires in 0.15s, chandelle builds over 0.50s. The aircraft gets better at every manoeuvre you practice, automatically.

17 Precision Flying

Precision flying on the MCO-1D works exactly as it does in a conventional aircraft. Set a constant throttle and attitude. The aircraft produces a consistent, predictable output. You read the instruments and make small corrections as needed. The aircraft does not make those corrections for you.

◆ **NOTE** The aircraft does not have an autopilot. It does not hold altitude, heading, or airspeed automatically. The reference bugs show you where the needles should be. You put them there and keep them there. This is flying.

What makes the instruments readable

A flapping wing at 1.87 Hz produces force that varies at twice the wingbeat frequency — 3.74 Hz. Without filtering, the VSI needle would oscillate ±82 ft/min just from the stroke cycle, making it unreadable. The VSI display averages over five wingbeats (~2.7 seconds), completely removing the stroke pulsation. What you see is the true mean vertical speed — the net effect of the aircraft and the air. This is the same principle as the pneumatic averaging in a conventional VSI instrument.

What makes the aircraft trimable

The precision focus directs the learning system to find wing configurations that produce minimum variation at neutral controls in calm air. After enough sorties in precision mode, the aircraft finds the cascade phase that produces near-zero yaw torque, the amplitude symmetry that produces near-zero roll drift, and the AoA that produces near-zero VSI at cruise lever. A well-trimmed aircraft requires only small corrections in calm air. The pilot makes those corrections.

NAS precision standards

VSI hold	±100 ft/min (±0.508 m/s) — standard IFR tolerance
Heading hold	±5° — standard IFR tolerance
Airspeed hold	±5 kt (±2.572 m/s) — standard IFR tolerance
Water approach VSI	±100 ft/min — same standard applies

How to fly to these standards

Set the reference bugs for your target values before entering the precision phase. Then fly the aircraft:

VSI: set lever and pitch until the VSI needle rests on the bug. Watch it. Correct with pitch when it drifts. Power for speed, pitch for rate.

Heading: set the heading bug before turning to the desired heading. Roll to put the compass card on the bug. Small rudder corrections for wind.

Altitude: set the altitude bug to your target. Watch the altimeter minor needle approach the triangle. Level off when the needle reaches the bug.

Airspeed: set lever. The aircraft produces consistent thrust at a given lever position. Airspeed then follows from your pitch attitude.

Water and seaplane approach

A water approach is standard precision flying to a 100 ft/min descent target. Set the VSI bug to -100, set power and pitch for that rate, fly it. The aircraft behaves identically over water and over land. There is no special water approach mode. Set the bug and fly.

"Target 100 ft/min descent / water approach / seaplane approach"
→ AI sets vsi_ref -100. Amber triangle appears at -100 on VSI gauge. Set your power and pitch to put the needle on it. You fly it.

Monitoring precision

"Precision status / how accurate is my altitude hold / VSI error"
→ AI reports all three bugs, current values, and errors. Also reports trim quality if controls have been neutral in calm air.

18 Forest Flight and Obstacle Avoidance

■ **CAUTION** Requires sensor ring hardware. Check with commissioning engineer.

"Going into the forest / enable obstacle avoidance"

→ Activates six time-of-flight sensor ring.

"The gap is X metres wide"

→ >0.55m: flapping passage 0.20-0.55m: sideways passage 0.12-0.20m: ballistic tuck

"Go through the gap"

→ Begins gap negotiation.

"Approach through the trees"

→ Canopy approach profile.

"Land here"

→ Vertical descent with position hold.

"Get out / climb out"

→ Confined exit.

19 Autonomous Learning Sorties

The aircraft flies, learns, and repeats. Define the area, start, swap batteries.

Class A (first 5)	Orientation — hover, cruise, rolls.
Class B (6-20)	Exploration — speed, climbs, precision cruise.
Class C (21+)	Performance — aerobatics, snap rolls, chandelles, top speed, water approach. Unlocks after Class B.

"Start the practice flights"

→ AI begins ALSS after checking fence.

"Stop the practice"

→ Ends after current sortie.

20 Ceiling Exploration

■ **IMPORTANT** May require regulatory authorisation. Obtain permissions first.

Two explicit steps required — never skip either.

"Arm ceiling exploration"
→ Step 1. Prints checklist. 60-second window opens.

"Confirm ceiling exploration (within 60s)"
→ Step 2. Probe begins at 0.15 m/s climb rate.

"Abort ceiling test"
→ Descends immediately.

21 After Flight

"We're done / land and save"

→ Saves all learning and disarms. One phrase does everything.

■ **CAUTION** Never disconnect the battery without saving learning first.

22 Troubleshooting

Interface OFFLINE

Check launcher running, aircraft light green, same network, ping 192.168.7.1.

Pre-flight fails on battery

Charge battery. Must be above 21.5V.

Pre-flight fails on GPS

Open area, away from buildings. Wait 60-90s.

Wing guard not calibrated

Measure fuselage width and TPU start. Ask AI "what measurements are needed".

VSI needle oscillating badly

This is unexpected — the 5-beat averaging should smooth the display. Check that the software version is current and the VSI telemetry key is vsi_smooth_fpm (not the raw vel_down_ms).

Aerobatic input not responding

Aircraft requires V > 8 m/s for aerobatic routing. Check airspeed. Urgency (how fast you throw the stick) selects snap vs aileron.

Cannot achieve precision tolerances

Say "precision flying" to engage the precision focus. Fly several sorties in calm air with controls neutral — the genome learns the trim configuration. Tolerances improve over sessions.

Lost connection during flight

Aircraft returns to takeoff autonomously. Do not approach until landed and powered down.

Appendix A Useful Phrases

Starting

"Ready to fly"
→ Pre-flight checks.

"Arm it"
→ Arms after one confirmation.

"We're done"
→ Saves learning, disarms.

Reference bugs

"Target 100 ft/min down / descent rate 100"
→ vsi_ref -100.

"500 ft/min descent"
→ vsi_ref -500.

"200 ft/min climb"
→ vsi_ref 200.

"Level flight / clear the VSI bug"
→ vsi_ref 0.

"Set heading bug to 090 / fly heading 090"
→ hdg_ref 090.

"Heading bug north / heading 360"
→ hdg_ref 360.

"Clear the heading bug"
→ hdg_ref_clear.

"Altitude bug 1000 feet"
→ alt_ref 1000.

"300 metres AGL"
→ alt_ref 300 m.

"Clear the altitude bug"
→ alt_ref_clear.

"Precision status / VSI error / how accurate"
→ precision_status.

Flight

"Take off"
→ Takeoff.

"Land"
→ Landing.

"Hold position"
→ GPS hold.

"Steep approach"
→ Lift dump.

Speed

"Top speed / as fast as possible"
→ top_speed focus.

"Get there quickly"
→ fast_cruise focus.

"Sprint"
→ sprint focus.

Climb

"Fastest climb"
→ best_climb_rate focus.

"Clear the trees"
→ best_climb_angle focus.

"J-hook"
→ Post-stall to hover (>20m).

Aerobatics

"Tight turns / aerobatics"
→ Aerobatics focus.

"That roll was good"
→ Records positive note.

Wing coordination

"Enable clap and fling"
→ Opens clap gate. ~25% hover lift.

"Anti-phase cruise"
→ 115° amplitude.

Precision

"Precision flying / fly steady / straight and level"
→ Precision focus.

"Water approach / seaplane approach"
→ vsi_ref -100. Fly it.

Forest flight

"Going into the forest"
→ Obstacle avoidance enabled.

"The gap is X metres wide"
→ Gap width set, strategy selected.

"Land here on the ledge"
→ Perch landing.

Learning

"How is learning going?"
→ Progress summary.

"Back to normal"
→ Clears focus.

ALSS

"Start practice flights"
→ Starts ALSS.

"Stop the practice"
→ Ends after current sortie.

"Abort / land now"
→ Immediate landing.

Ceiling

"Arm ceiling exploration"
→ Step 1.

"Confirm ceiling exploration"
→ Step 2 — begins probe.

"Abort ceiling test"
→ Descends immediately.

Appendix B What the Aircraft Says

"All checks pass. Ready to arm."
Pre-flight passed.

"Arming now."
Stand clear of wings.

"Alert: fault detected."
Ask "what is the fault?" immediately.

"Alert: battery low (X.Xv)."
Plan to land. Below 20V: land now.

"Wing guard not calibrated — fuselage width needed."
Measure spar-to-spar. Report with: GS wc_set_fuselage_width [mm]

"Clap gate open — dorsal mode."
Clap-and-fling active. 15-25% hover lift. Guard still validates every tick.

"Gap ahead ~Xm wide. Say "go through the gap" to enter."
Forward sensor detected passable gap. Say the words to proceed.

"ALSS Sortie N complete. Swap battery and re-arm."
Swap battery, arm again, next sortie begins.

"Session saved. Safe to disconnect."
All learning saved. Safe to cut battery.

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Revision 23 — August 2026

New in v23: Standard six-pack instrument layout (§4), Reference bugs for VSI / heading / altitude (§5), Precision flying section (§17). The aircraft does not hold altitude, heading, or airspeed automatically. The bugs are reminders. The pilot flies.